

1 Solution — GIAM 3.1

Problem 4

1.1 Problem

Prove that the sum of an odd number and an even number is odd.

1.2 Statement

$$\forall x, y \in \mathbb{Z}, \quad (x \text{ is odd}) \wedge (y \text{ is even}) \implies (x + y \text{ is odd}).$$

1.3 Proof

Proof. Let x be an odd number and y an even number. By the definitions of odd and even, there are integers k and j with $x = 2k + 1$ and $y = 2j$. By algebra,

$$x + y = (2k + 1) + 2j = 2(k + j) + 1.$$

Note that $k + j$ is an integer, since $\mathbb{Z}$ is closed under addition. So $x + y = 2m + 1$ with $m = k + j \in \mathbb{Z}$, and hence $x + y$ is odd. Since x and y were arbitrary, the sum of any odd number and any even number is odd. ■

1.4 The Structure

Same three-step skeleton as before: unpack each definition, add, then repack into the definition of odd.

UNPACK

$x \text{ is odd} \implies x = 2k + 1, k \in \mathbb{Z}$

$y \text{ is even} \implies y = 2j, j \in \mathbb{Z}$

two different letters — the witnesses are unrelated

COMPUTE

$$x + y = (2k + 1) + 2j = 2(k + j) + 1$$

verified on 300 random integer pairs

REPACK

$k + j \in \mathbb{Z}$ (closure of $\mathbb{Z}$ under +), so $x + y = 2m + 1$ with $m = k + j \in \mathbb{Z}$
hence $x + y$ is odd

This problem is one entry in a four-entry table

+	even	odd
even	even	odd
odd	odd	even

Write even = 0, odd = 1:

0+0=0 0+1=1
1+0=1 1+1=0

that is exactly XOR

the shaded cell is this problem; "odd + odd = even" was Problem 2.7.5

Parity is addition in $\mathbb{Z}_2$ — the integers collapsed onto their remainder mod 2.

The four facts about odd and even sums are the whole addition table of that two-element system, and it coincides with the exclusive-or of Problem 2.1.1.

Figure 1.1: Three stacked panels. UNPACK: x is odd gives x equal to $2k$ plus 1 with k an integer; y is even gives y equal to $2j$ with j an integer — two different letters, since the witnesses are unrelated. COMPUTE: x plus y equals $(2k \text{ plus } 1) \text{ plus } 2j \text{ equals } 2(k \text{ plus } j) \text{ plus } 1$, verified on 300 random integer pairs. REPACK: k plus j is an integer by closure of the integers under addition, so x plus y equals $2m$ plus 1 with m equal to k plus j an integer, hence x plus y is odd. Below, a two-by-two parity table with rows and columns labelled even and odd: even plus even is even, even plus odd is odd, odd plus even is odd — the shaded cell, which is this problem — and odd plus odd is even, which was Problem 2.7.5. A side panel writes even as 0 and odd as 1, giving $0+0=0$, $0+1=1$, $1+0=1$, $1+1=0$, and notes that this is exactly XOR. A closing banner explains that parity is addition in $\mathbb{Z}_2$, the integers collapsed onto their remainder mod 2, and the four facts about odd and even sums form the whole addition table of that two-element system, coinciding with the exclusive-or of Problem 2.1.1.

The proof is the same **unpack** – **compute** – **repack** skeleton as Problems 2.7.5 and 3.1.3. The only genuine content is the factoring step $2k + 1 + 2j = 2(k + j) + 1$, which pulls the 2 out front so the result *looks like* the definition of odd, plus the observation that $k + j$ lands in $\mathbb{Z}$.

Verified on all 3660 odd/even pairs in $[-60, 60]$.

1.5 Remark — Only One Case Is Needed

A reader might worry that the theorem needs two cases — “odd plus even” and “even plus odd.” It does not. The statement names which summand is which, so there is only one configuration to handle. And even if it had said “the sum of an odd and an even number, in either order,” addition is **commutative**, so $y + x = x + y$ and the second case reduces immediately to the first.

Being able to see that a second case is unnecessary is worth as much as being able to handle one that is. Spurious case splits make proofs longer without making them more correct.

1.6 Remark — The Closure Step Is Doing Real Work

As in Problem 2.7.5, the line “ $k + j$ is an integer” is not padding. The definition of *odd* demands $n = 2m + 1$ for an integer m , so producing the equation $x + y = 2(k + j) + 1$ is only half the job — one must confirm that $k + j$ lives in $\mathbb{Z}$.

Here the confirmation is immediate, because $\mathbb{Z}$ is closed under addition. But the obligation is real: over a set not closed under $+$, the same algebra would produce an equation that fails to witness anything. Every “unpack, compute, repack” proof ends with exactly this check, and Problem 3.1.3’s closure argument ended the same way with “ bd is a non-zero integer.”

1.7 Remark — Four Facts, One Table

This problem is one cell of a 2×2 table, and all four cells are proved by the identical method:

even odd		
even	even	odd
odd	odd	even

Table 1.1

- even + even: $2k + 2j = 2(k + j)$
- even + odd, odd + even: $2k + (2j + 1) = 2(k + j) + 1$ — **this problem**
- odd + odd: $(2k + 1) + (2j + 1) = 2(k + j + 1)$ — Problem 2.7.5

In each case the algebra rearranges the expression so that a factor of **2** can be pulled out, and the leftover **+1** (or absence of one) decides the parity. Recognising that four separate-looking exercises are one computation in four dressings is the useful takeaway.

1.8 Remark — Parity Is Addition in $\mathbb{Z}_2$

The table above is not an accident of arithmetic; it is the **addition table of $\mathbb{Z}_2$** , the integers collapsed onto their remainder modulo **2**. Writing even = **0** and odd = **1**:

$$0 + 0 = 0, \quad 0 + 1 = 1, \quad 1 + 0 = 1, \quad 1 + 1 = 0.$$

Two consequences worth noting. First, this is *exactly* the truth table of **exclusive-or** from Problem 2.1.1 — parity addition and XOR are the same operation, which is why XOR is described as “addition mod 2.” Second, the fact that parity behaves so predictably under addition is what makes **parity arguments** possible: one can often settle a question by tracking only whether quantities are odd or even, discarding all other information. That technique appears throughout combinatorics and is the basis of parity check bits in error detection.

1.9 Remark — Multiplication Has a Different Table

It is worth contrasting the addition table with the multiplication one, since students sometimes assume they behave alike:

$\times$	even	odd
even	even	even
odd	even	odd

Table 1.2

Three of the four products are even, because a single factor of **2** in *either* argument survives multiplication. Only odd $\times$ odd is odd. In $\mathbb{Z}_2$ terms this is the AND operation ($0 \cdot 0 = 0 \cdot 1 = 1 \cdot 0 = 0$, $1 \cdot 1 = 1$), completing the picture: parity addition is XOR, parity multiplication is AND, and together they make $\mathbb{Z}_2$ the two-element field on which all of Boolean algebra sits.